# Ananya Kulkarni

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## PROFESSIONAL SUMMARY

Results-driven software engineer with strong foundation in distributed systems, high-concurrency backend services, and clean architecture. Experienc

## EDUCATION

**Bachelor of Technology in Computer Science & Engineering — Vellore Institute of Technology** *2020 - 2024*

Cumulative Grade Point Average: 9.80/10.0

## WORK EXPERIENCE

**Platform Engineer — PhonePe Pvt. Ltd., Bengaluru, India** *Jun 2023 - Present*

- Implemented automated circuit-breaker pattern with Resilience4j across 8 downstream microservices, preventing cascade outages during third-pa
- Engineered distributed idempotency layer with Redis and PostgreSQL, eliminating 100% of double-debit anomalies during peak flash sales.
- Architected high-throughput payment webhook processor using Go and Redis Streams, handling 14,000 requests/sec with p99 latency under 28m

## TECHNICAL PROJECTS

**Autonomous Code Review Linter & Security Auditor | Technologies: Python, AST, Tree-sitter, Docker, FastAPI**

- Engineered static analysis pipeline parsing abstract syntax trees (ASTs) to detect SQL injection, unvalidated deserialization, and memory leaks.
- Built automated GitHub Action integration running checks across 200+ pull requests with an average scan duration under 3.5 seconds.
- Containerized execution sandbox using gVisor and Docker to securely isolate untrusted student code submissions.

**Distributed Log-Structured Key-Value Store | Technologies: Go, Raft Consensus, gRPC, LSM-Tree, Docker**

- Implemented distributed LSM-tree storage engine in Go with write-ahead logging (WAL), SSTable compaction, and bloom filters.
- Embedded Raft consensus algorithm for distributed multi-node replication, surviving arbitrary node crashes without state divergence.
- Benchmarked performance using custom concurrent load generator, achieving 85,000 read operations/sec across a 5-node cluster.

## TECHNICAL SKILLS

Languages: Go, Java, Python, TypeScript, C++, SQL

Frameworks & Systems: Spring Boot, Node.js, FastAPI, React, Next.js, gRPC

Databases & Storage: PostgreSQL, Redis, MongoDB, Apache Kafka, Elasticsearch

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, S3, RDS), Git, CI/CD Pipelines